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Vortex86DX-9121D Modification List

2009/06/12

Now the final version of Vortex86DX is fully tested and formally released

The item number is
- Vx86DX-9121D (SoC only)
- Vx86DX-9121D-BS (SoC with AMI BIOS License Lable)

You can also see the top marking as below.



Please see the modification from previous versions as below.

9120A shuttle chip Bug

- ODT fail, DDRII load mode need extra delay(after wait DDRII PHY lock) for 2nd bank.
- HLT instruction can stop CPU clock
- SPI_DO had output when do SPI read
- Jtag enable causing CPU wrong on DMP's PCB
- SD register busy state
- USB touch panel issue (Low speed dword align issue)
- UART TXD_EN not drive high in Stop bit time
- ISA BUS compatibility, IOW MEMW (Low go High) , data bus hold data for less than 100ns , the old ISA design is can not latch the data. Should change to 150 ns.
- RS232 IRQ overflow (by Keyboard IRQ1, Temporary solution is modify the BIOS). MP version bug fixed by UART FIFO extend to 32byte

9120B others modification list:

L2 cache: 128KB => 256KB, direct map => 4 ways

- Add DDR2 power saving
- Changed DQS0+/- DQS1+/- ball-out
  A6: DQS#[0], A7: DQS[0]
  D14: DQS#[1], E16: DQS[1]
- IDE use standard register define

9121C modification list:

- IF IDE decode of the channel is disabled, R/W the I/O,D1011 always target retry.
- Secondary IDE strap pin disabled, the Data bit 7 always is
- UART TX add a Idle bit.
- LPC DMA ready signal did not response during floppy formatting.

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